Is it chance?

Calibrating and tuning the group identification

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Method note, sections 5–7: `xc-lab/ideas/group-identification`

The question

Same site, same ridges, same house thermals: pilots meet without interacting.

$$N(\tau) = \#\{\text{real contacts} \geq \tau\}, \qquad C(\tau) = \text{expected by chance}$$

$$\text{keep } \tau \geq \tau^*, \qquad \frac{C(\tau^*)}{N(\tau^*)} \leq q$$

162 485 flights, 6 209 dates, FFVL archive; profiles on 300 dates.

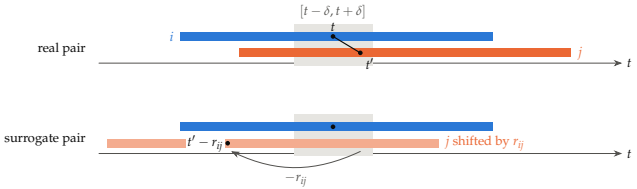
Part 1: chance

Null model, false discovery rate, τ^*

Null model: shift one flight in time

Surrogate pair. i, j same date, $\|\mathbf{x}_i(k_i^0) - \mathbf{x}_j(k_j^0)\| < R_s = 10$ km:

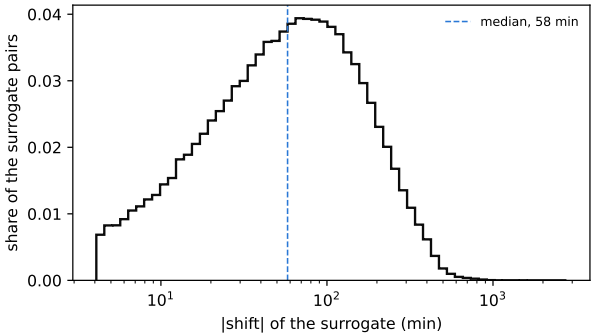
$$\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r), \qquad r = (k_j^0 - k_i^0) - \vartheta, \qquad \vartheta \sim \hat{P}(k_j^0 - k_i^0), \qquad |r| \geq 2\delta + \Delta$$



$$(t, t') \in \mathcal{M}_{ij} \;\Rightarrow\; |t - (t' - r)| \geq |r| - \delta > \delta$$

Keeps route, terrain, date; destroys every real correspondence. Spiegel et al. (2016).

Shifts of about an hour



248 741 surrogate pairs, whole archive.

$$\text{median } |r| = 58 \text{ min}$$

$$|r| \in [8 \text{ min}, 4.4 \text{ h}] \quad (90\%)$$

$\Rightarrow$ the copy flies in the weather of another hour

False discovery rate

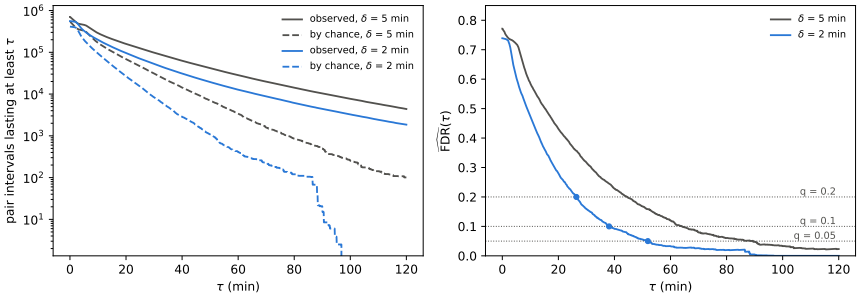
Date d. $M_d(\tau)$ surrogate intervals $\geq \tau$; S_d, H_d surrogate and real exposure (time both fly).

$$\rho_{0,d}(\tau) = \frac{M_d(\tau)}{S_d}, \qquad C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d$$

$$\widehat{\mathrm{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\{\tau : \widehat{\mathrm{FDR}}(\tau) \leq q\}$$

SAM (Tusher et al. 2001), with time shifts in place of permutations.

Most short contacts are chance



$$\widehat{\mathrm{FDR}}(0) = 0.74, \qquad \widehat{\mathrm{FDR}}(10\ \mathrm{min}) = 0.47, \qquad \widehat{\mathrm{FDR}}(30\ \mathrm{min}) = 0.16$$

$$q = 0.1 \ \Rightarrow \ \tau^* = 38.1\ \mathrm{min} \quad [37.0,\ 41.1]$$

$\delta = 2$ min (blue), 5 min (grey). Bracket: 5–95% over 200 resamplings of the dates.

How much the shift matters

surrogates	$\widehat{\mathrm{FDR}}(10\text{ min})$	$\widehat{\mathrm{FDR}}(30\text{ min})$	τ^* (min)
all $\lvert r\rvert$	0.48	0.16	38.2
$\lvert r\rvert < 30$ min	0.73	0.31	53.3
$\lvert r\rvert \geq 60$ min	0.26	0.05	23.0

$\tau^* \in [23, 53]$ min, set by $|r|$

short $|r|$: same weather, but also followers a few minutes apart

$q = 0.1$, $\delta = 2$ min, 2 737 dates with all classes. We keep r from the takeoff delays.

Part 2: parameters

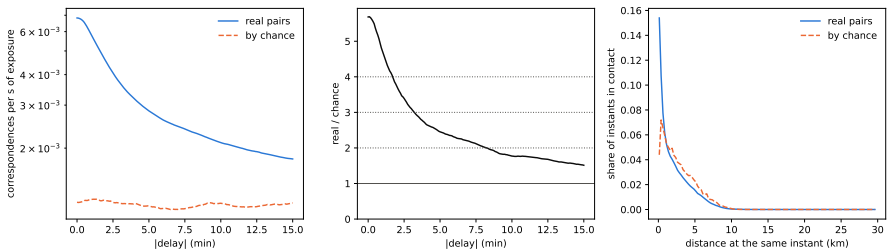
Measured, chosen, or definitions

Three kinds of parameters

	value	set by
δ	2 min	measured: $R(l)$
q	0.1	chosen
τ^*	38.1 min	$\widehat{\mathrm{FDR}}(\tau^*) \leq q$
ε_h, ε_z, g	500 m, 300 m, 120 s	definitions: no edge
$\theta_{\max}$	45°	no effect
Δ, $h_{\max}$, R_{dep}, $v_{\max}$, R_{s}, $r_{\min}$		fixed: bounds, exactness

δ: delay profile

$$R(l) = \frac{n_{\text{real}}(l)/H_{\text{real}}}{n_{\text{chance}}(l)/H_{\text{chance}}}, \qquad n(l) = \#\{(t, t+l) \in \mathcal{M}\}$$



$$R_{\max} = 5.6, \quad R(105\,\text{s}) \approx 4, \quad R(195\,\text{s}) \approx 3, \quad R(500\,\text{s}) \approx 2, \quad R(15\,\text{min}) = 1.5$$

300 dates, window 15 min. Fast part: flying together; slow tail: same thermals, same hour.

δ: long contacts, short delays

$$\ell_{\text{med}} = \text{median}\{l^*(t) : \; t \in [t_{\text{on}}, t_{\text{off}}), \text{ no search}\}$$

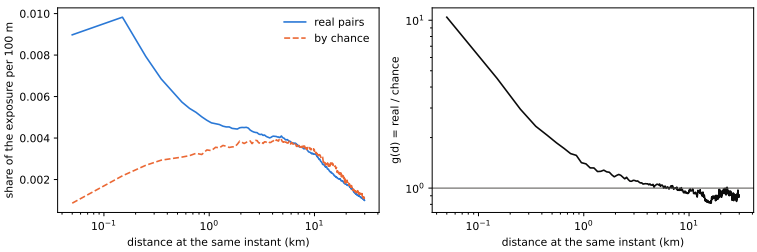
duration	median $\lvert\ell_{\text{med}}\rvert$	$\lvert\ell_{\text{med}}\rvert \le 2$ min	$\lvert\ell_{\text{med}}\rvert \ge 285$ s
<10 min	210 s	32%	29%
10–30 min	160 s	39%	12%
30–90 min	80 s	64%	2.1%
≥ 90 min	25 s	90%	6 of 10 734

$$\delta = 2 \text{ min}: \qquad R(\delta) = 3.7, \qquad P\big(|\ell_{\text{med}}| \le \delta \;\big|\; \text{duration} \ge 90 \text{ min}\big) = 0.9$$

Run with $\delta = 5$ min, 965 770 intervals.

Distance at the same instant

$$g(d) = \frac{p_{\text{real}}(d)}{p_{\text{chance}}(d)}, \qquad d = \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|$$

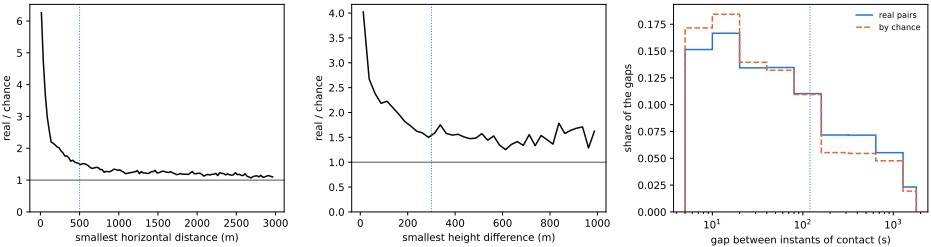


$g(d < 100\text{ m}) = 10.4, \quad g < 1 \text{ for } d > 5.7\text{ km}, \quad \text{in contact: } d \leq \varepsilon_h + v_{\max}\delta \approx 3.8\text{ km}$

210 dates, every instant both fly. Excess beyond sight (1.5 km): weather of the hour too.

Tolerances: no edge

$$d_h(t) = \min_{|l|\le\delta,\ |\Delta z_l|<\varepsilon_z} \|\Delta \mathbf{x}_l\|, \qquad d_z(t) = \min_{|l|\le\delta,\ \|\Delta \mathbf{x}_l\|<\varepsilon_h} |\Delta z_l|$$



$$\frac{\text{real}}{\text{chance}} \approx 1.5 \text{ at } \varepsilon_h,\ \varepsilon_z; \qquad \text{gaps: same lengths } \Rightarrow\ \varepsilon_h, \varepsilon_z, g \text{ definitions}$$

$\Delta \mathbf{x}_l = \mathbf{x}_i(t) - \mathbf{x}_j(t+l)$, $\Delta z_l = z_i(t) - z_j(t+l)$; 210 dates, $\delta = 2$ min.

One parameter at a time: no plateau

$$f_G = \frac{\sum_i \left| \{ t \in T_i^{\text{dep}} : \ \gamma_i(t) \notin \{L, S\} \} \right|}{\sum_i |T_i^{\text{dep}}|}$$

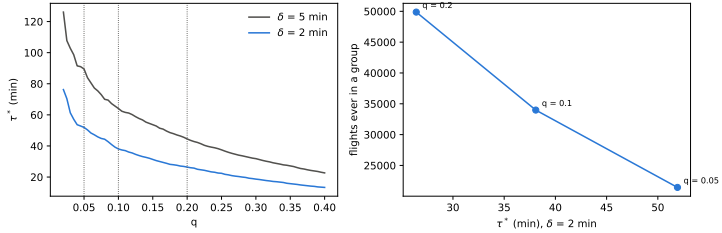
setting	τ^* (min)	f_G	episodes
start	126.1	0.034	98
$\varepsilon_h = 250$, 1000 m	75.1, 126.8	0.058, 0.065	253, 162
$\varepsilon_z = 150$, 600 m	126.1, 126.1	0.020, 0.044	61, 122
$\delta = 2$, 10 min	42.6, 128.4	0.106, 0.062	600, 153
$\theta_{\max} = 30°, 90°$	126.1, 126.1	0.034, 0.034	97, 98
$g = 60$, 300 s	75.8, 126.1	0.087, 0.055	332, 141

300 dates, start $\delta = 5$ min, $q = 0.05$, τ^* recalibrated each time.

$\theta_{\max}$: f_G unchanged

others: $f_G \times \leq 3$, via τ^*

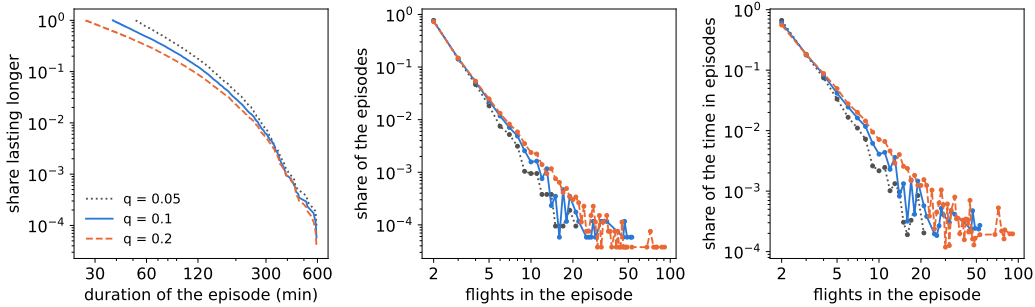
q: how much chance we accept



q	τ^* (min)	para in a group	hang in a group	episodes
0.05	51.9	13.7%	3.8%	10 553
0.1	38.1	21.7%	6.9%	17 169
0.2	26.4	31.7%	13.9%	26 504

$\delta = 2$ min. We take $q = 0.1$.

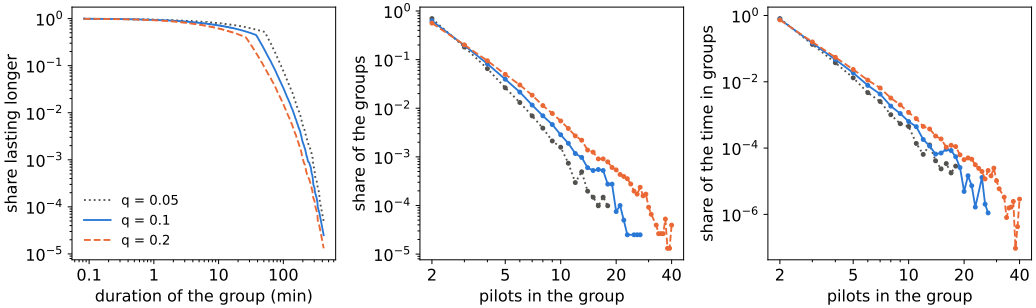
q: episodes, how long and how many flights



Whole archive, $\delta = 2$ min, $q = 0.05, 0.1, 0.2$. Left: share lasting longer. Middle: share of the episodes of each size. Right: share of the episode time. Size: distinct flights of the episode.

With $q = 0.1$ the median episode lasts 58 min and 74% are pairs. Weighted by time the pairs drop to 61%: large episodes last longer.

q: groups, how long and how many pilots at once



Whole archive, $\delta = 2$ min, $q = 0.05, 0.1, 0.2$. A group keeps its members; each arrival or departure starts a new one. Size: pilots at once.

With $q = 0.1$ the median group lasts 30 min and 63% are pairs. Weighted by time the pairs rise to 77%: large groups change quickly.

Limits

- chance = null model: $|r| \sim 1$ h removes the weather of the hour $\Rightarrow$ "real" includes same thermal, same time
- $\tau^* \in [23, 53]$ min with the class of $|r|$
- profiles: 300 dates; plateau: 300 dates, $\delta = 5$ min
- data: old repository; phases on 1 Hz flights only

In short

$$\text{chance:} \quad \mathbf{x}_j(t) \to \mathbf{x}_j(t+r), \quad |r| \sim 1 \text{ h}$$

$$\widehat{\text{FDR}}(\tau^*) \leq 0.1 \;\Rightarrow\; \tau^* = 38.1 \text{ min}$$

$$\delta = 2 \text{ min}: \quad R(\delta) = 3.7, \quad 90\% \text{ of long contacts}$$

$$\varepsilon_h,\ \varepsilon_z,\ g: \text{ definitions}, \quad f_G \times \leq 3$$

In a group at some point: 21.7% of the paraglider flights, 6.9% of the hang-glider flights; 17 169 episodes.